

Dimerization and dynamics of angiotensin-I converting enzyme revealed by cryoEM and MD simulations

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eLife Assessment

This **important** study shows, for the first time, the structure and snapshots of the dynamics of the full-length soluble Angiotensin-I converting enzyme dimer. The combination of structural and computational approaches elucidates with **convincing** evidence the conformational dynamics of the complex and key regions mediating the conformational change. This work provides an example of how conformational heterogeneity can be used to gain insights into protein function.

<https://doi.org/10.7554/eLife.106044.1.sa3>

Abstract

Angiotensin-I converting enzyme (ACE) regulates the levels of disparate bioactive peptides, notably converting angiotensin-I to angiotensin-II and degrading amyloid beta. ACE is a heavily glycosylated dimer, containing 4 analogous catalytic sites, and exists in membrane bound and soluble (sACE) forms. ACE inhibition is a frontline, FDA-approved, therapy for cardiovascular diseases yet is associated with significant side effects, including higher rates of lung cancer. To date, structural studies have been confined to individual domains or partially denatured cryoEM structures. Here we report the cryoEM structure of the full-length, glycosylated, sACE dimer. We resolved four structural states at 2.99 to 3.65 Å resolution which are primarily differentiated by varying degrees of solvent accessibility to the active sites and reveal the full dimerization interface. We also employed all-atom molecular dynamics (MD) simulations and heterogeneity analysis in cryoSPARC, cryoDRGN, and RECOVAR to elucidate the conformational dynamics of sACE and identify key regions mediating conformational change. We identify differences in the mechanisms governing the conformational dynamics of individual domains that have implications for the design of domain-specific sACE modulators.

Introduction

Angiotensin-I converting enzyme (ACE) is an M2 clan zinc metalloprotease (EC3.4.15.1) located on the plasma membrane and in the extracellular milieu (1–3). ACE plays a crucial role in the renin-angiotensin system by converting angiotensin I to the potent vasoconstrictor angiotensin II, and in the kinin-kallikrein system by degrading bradykinin, a potent vasodilator. Inhibiting ACE is a primary approach for treating hypertension and is a proven therapy for heart failure, diabetic nephropathy, and other cardiovascular and renal dysfunctions (4). However, ACE inhibition has known side effects, e.g., dry cough and angioedema, and has been linked to higher rates of lung cancer (Wu *et al.*, 2023). This is partly because ACE also degrades a diverse range of peptides, resulting in complex physiological outcomes due to the interplay in altering levels of these peptides (4–8). Furthermore, ACE degrades amyloid β (A β), which is associated with the progression of Alzheimer's diseases (9–12). Overexpression of ACE in myelomonocytes substantially reduced A β load and prevented Alzheimer's disease-like cognitive decline in mice (13). However, the brains of some Alzheimer's disease patients showed upregulation of the renin-angiotensin system which led to deleterious effects and ACE inhibition was beneficial (14–16). These studies indicate that global ACE inhibition could be either beneficial or harmful for individual Alzheimer's patients, and underscores a need for better understanding the molecular mechanisms governing ACE function and substrate selectivity to develop more selective, improved, ACE inhibitors and expand ACE-based therapies (14–16).

ACE is a 180 kDa heavily N-linked glycosylated type 1 transmembrane protein with two homologous catalytic domains residing at the N- and C-terminal ends (ACE-N and ACE-C) in its extracellular region (1, 3, 17). The ACE extracellular region can be released via proteolytic cleavage near the transmembrane region by ADAM family protease(s), generating soluble ACE (sACE) (18). ACE belongs to the cowrin (cowry like) family of zinc metallocarboxypeptidases because its catalytic domain is ellipsoidal and contains a long, deep, yet narrow catalytic cleft (19). ACE is known to cleave short peptides, such as angiotensin I (decapeptide) and bradykinin (nonapeptide) by removing the C-terminal two amino acid residues. Different from the funnelin family of metallocarboxypeptidases that contains a funnel-like cavity, the catalytic cleft of ACE can accommodate entire oligopeptides and ACE has been shown to cleave multiple discrete sites on A β outside the C-terminal end (11). Thus, it is more accurate to refer to ACE as an endopeptidase with dipeptidyl carboxylase activity.

Extensive crystallographic studies of ACE-N or ACE-C alone and in complex with various inhibitors and a catalytic product, angiotensin-II, have elucidated the overall structures of ACE catalytic domains and molecular details of the catalytic pockets (2, 20–23). The structures indicate that each domain is comprised of three sub-domains and offer insight for the substrate preferences around cleavage sites and the promiscuity of each domain toward short peptide substrates. Furthermore, they reveal the orientation of substrate binding sites around scissile bond, indicating that the entire oligopeptide substrates need to be engulfed into the catalytic chamber of ACE catalytic domain (20, 22, 23).

However, the catalytic pocket of ACE in the closed state is not large enough to accommodate the entrance of all its known substrates, particularly A β , to permit the observed cleavage pattern. Thus, similar to other A β -degrading proteases, such as insulin degrading enzyme (IDE), presequence protease (PreP), and neprilysin (NEP), ACE must undergo a large-scale open-closed transition to capture and degrade larger peptide substrates such as A β (24–26). Recent open state structures of sACE-N, sACE monomer, and a sACE-N dimer, along with molecular dynamics (MD) simulations of sACE-C, have begun to reveal the conformational heterogeneity, though it remains under-studied (27–29).

The extracellular domains of ACE are readily dimerized in the membrane-bound and soluble forms. Both the ACE-N and ACE-C domains within the ACE monomer are catalytically active and each has its own substrate preferences (5 [↗](#), 11 [↗](#), 30 [↗](#)). For example, ACE-N can convert A β 1-42 to A β 1-40, which is less amyloid fibril-prone than A β 1-42 but ACE-C cannot (11 [↗](#)). Accumulating evidence support a complicated interplay between ACE-N and ACE-C within the ACE dimer. Enzyme kinetics analysis suggests negative cooperativity between two catalytic domains (31 [↗](#)–33 [↗](#)). However, ACE also exhibits positive synergy toward A β cleavage and allostery to enhance the activity of its binding partner, bradykinin receptor (11 [↗](#), 34 [↗](#)). Recently, cryoEM structures of soluble dimeric apo-sACE were reported (28 [↗](#)); however, due to the poor resolution/denaturation of sACE-C, only full-length sACE monomer or dimeric sACE-N was reported yet the conformations preclude full-length sACE dimerization due to the severe steric clash between sACE-C domains. Here, we present four discrete cryoEM structures of full-length sACE that contain all four catalytic domains (**Fig. 1A** [↗](#)). This is followed by cryoEM heterogeneity analysis and all-atom MD simulations. Together, our analyses provide structural insights into domain interactions, open-closed transitions of the catalytic domain, and conformational dynamics of soluble dimeric ACE.

Results

Structures of the full-length sACE dimer determined by single particle cryoEM

Soluble human ACE (sACE), which includes the entire extracellular region of ACE, was expressed in HEK293F cells and purified from culture supernatant using an ion exchanger (Source Q) followed by size exclusion chromatography (Superdex S200). Buffer conditions suitable for single particle cryoEM analysis were identified and optimized using differential scanning fluorimetry (Figure S1). Purified sACE was then vitrified using a Vitrobot, and a dataset of ~3,600 micrographs was collected on a 300 keV Titan Krios equipped with a Gatan K3 camera at the University of Chicago Advanced Electron Microscopy Facility (Table S1). The data was processed in cryoSPARC v4.4.0 (Figure S2). After filtering ~1.1 million picked particles to ~378K using 2D and 3D classification, three classes were identified (**Figure 1** [↗](#), Figure S2). Of these, ~268K particles mapped to a dominant 2-domain class reconstructed as an ACE-N dimer, while ~21K particles mapped to a minor 3-domain class containing two ACE-N and only one ACE-C, reminiscent of the recently published ACE cryoEM structure (28 [↗](#)). The remaining 89K particles mapped to a full-length ACE dimer containing two ACE-N and two ACE-C, which was refined to 3.65Å resolution (**Figure 1C** [↗](#), S2, Movie S1, Table S1).

The partial denaturation of macromolecules during vitrification has been attributed mainly to repeated exposure of the macromolecules to the air-water interface (35 [↗](#), 36 [↗](#)). The sACE-C domain was shown to be less stable than the ACE-N domain (37 [↗](#)), so we hypothesized that repeated exposure to the air-water interface could cause preferential denaturation of ACE-C. We have previously showed that preferential denaturation of human PreP occurred in one of its two homologous domains and that faster vitrification using Chameleon significantly prevented such preferential denaturation (24 [↗](#)). Therefore, we collaborated with the National Center for CryoEM Access and Training (NCCAT) to prepare sACE grids using Chameleon (38 [↗](#)). Chameleon preparation reduced the time the sample spends on the grid before freezing by more than 10-fold. We collected a new dataset of ~18,000 micrographs and our 2D classification revealed a higher percentage of particles for four-domain classes than those prepared using the Vitrobot (**Figure 1B** [↗](#)). However, our 3D classification still showed the existence of structures of the ACE-N only dimer and those containing two ACE-N and one ACE-C (Figure S3). This indicates that the reduction in vitrification time using Chameleon reduced but did not prevent preferential denaturation.

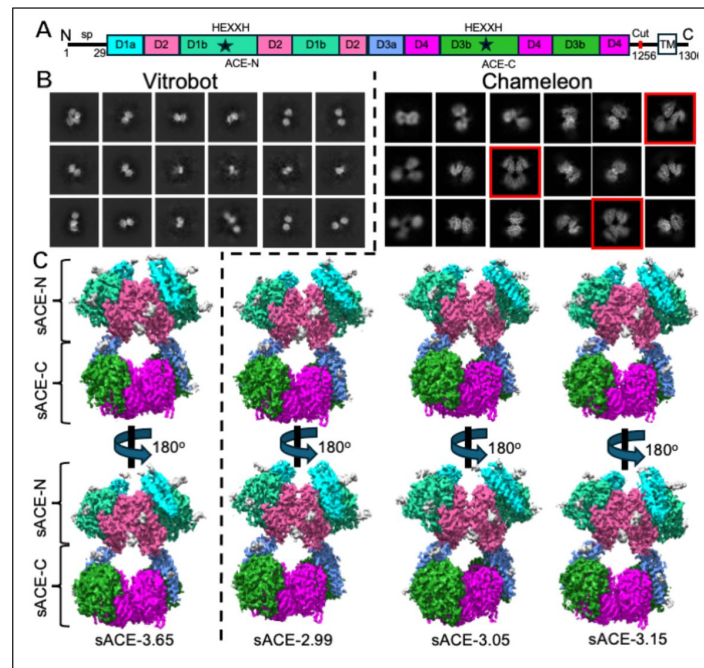


Figure 1

CryoEM analysis of human ACE dimer.

(A) Diagram for the key features of domain on primary sequence of human ACE. To avoid confusion existing in the numbering of ACE structures, we number ACE based on Uniprot P12821-1. D1a and D3a domains, also called "lid" encompasses residues 31-127 and aa 645-725, respectively. D1b and D3b domains each have two discrete segments; residues 292-465 and 525-603 for D1b and residues 897-1060 and 1120-1200 for D3b. D2 encompasses three discrete segments, residues 128-291, 466-524, and 604-644 while D4 domain encompasses residues 726-896, 1061-1119, and 1201-1231. sp = signal peptide. asterisk = zinc binding motif. The construct used in this study comprises residues 1-1231, which we refer to as the soluble region of ACE (sACE). Colored by sub-domain: D1a cyan, D1b aqua, D2 pink, D3a light blue, D3b green, D4 magenta. (B) 2D classification of human sACE particles from grids made by vitrobot and Chameleon. Clear four domain classes visible in the Chameleon-derived classification are boxed in red, similar views are lacking in the vitrobot dataset. (C) Full-length sACE 3D volumes, colored by sub-domain as in (A). Glycan density is shown in gray. See Figure S2 for vitrobot-prepared data processing details, Figure S3 for Chameleon-prepared data processing details and Table S1 for data refinement statistics.

From these data, we obtained a map for a consensus ACE structure that achieved a resolution of 2.8Å. However, while the core region showed excellent map quality, the map quality for the dynamic regions corresponding to the open-close transition of ACE catalytic domains was poor, suggesting multiple discrete conformational states exist. We thus performed 3D classification and reconstructed three distinct conformational states of full-length apo-ACE at resolutions of 2.99Å, 3.05Å, and 3.15Å (**Figure 1C** [↗](#), S3, Movie S2-4, Table S1). We refer to these ACE dimer structures by their resolutions, namely ACE-2.99, ACE-3.05, and ACE-3.15, along with the structure from data obtained using the Vitrobot, ACE-3.65. Given the rich structural information from more than 50 reported crystal structures of ACE and one recently reported cryoEM study of ACE, below we focus on the new insights that our ACE dimer structures offer ([2](#) [↗](#), [20](#) [↗](#), [22](#) [↗](#), [28](#) [↗](#), [39](#) [↗](#), [40](#) [↗](#)).

The overall structures of sACE dimers and structural insight into dimerization

Soluble ACE has two homologous catalytic domains: ACE-N (amino acids 31-644) and ACE-C (amino acids 645-1231). Based on the comparison of various catalytic domains of our ACE dimer structures, we subdivide each catalytic domain of ACE into three subdomains: D1a, D1b, and D2 for ACE-N, and D3a, D3b, and D4 for ACE-C (**Figure 1A** [↗](#)). These three subdomains in either ACE-N or ACE-C form a catalytic chamber. The catalytic zinc ion is coordinated by the canonical HEXXH motif residing in the D1b and D3b subdomains (**Figure 1A** [↗](#)). Four sACE dimer structures manifest as pseudo-symmetric C2 homodimers across the dimer interface. The dominant symmetry-breaking features are the degrees of displacement between D1 and D2 or between D3 and D4, leading to changes in solvent accessibility to the catalytic pockets.

Based on the openness of the catalytic chamber, defined at the distance between the D1/3a and D2/4 helices that gate access, we classified sACE-N into three conformational states: open (O), intermediate open (I), and closed (C) (**Figure 2A** [↗](#)); sACE-C is significantly less open than sACE-N in our structures, and is only classified into intermediate (I) and closed (C) states (**Figure 2B** [↗](#)). The four sACE dimer structures differ mostly by the degrees of openness of their four catalytic domains (**Figure 2C, D** [↗](#)). Our sACE-2.99 and sACE-3.65 structures are globally similar, both contain sACE-N domains in the open and intermediate states and sACE-C domains in the closed and intermediate states. In sACE-3.05, both sACE-N domains adopt an open state, while the sACE-C domains adopt an intermediate and closed state. In the sACE-3.15 structure, one subunit has both sACE-N and sACE-C adopting a closed conformation, while the other subunit has an open sACE-N domain and an intermediate sACE-C domain.

Our four cryoEM structures of sACE provide the structural basis for sACE dimerization. sACE dimerization is mediated by interfaces between D2 subdomains and D4 subdomains of sACE monomers. The D2 and D4 subdomains generally have much lower B-factors than the D1 and D3 subdomains (Figure S4). The interaction of the D4-D4 subdomains is analogous to the D2-D2 interaction, yielding an RMSD of $\sim 1 \text{ Å}^2$ when aligned (**Figure 3A** [↗](#)). The primary source of variation is a general rigid body shift away from the interaction interface for non-interacting regions in the D4-D4 region compared to the D2-D2 region. The observed D2-D2 interface is consistent with one of three possible contacts between sACE-N in the non-crystallographic symmetry mates from previous crystallographic analysis of sACE-N and the previously published cryoEM structure (Figure S5). In total, the interface between sACE monomers buries a surface area of approximately 1,900 Å². Although sACE-N and sACE-C are structurally analogous, their respective interfaces (sACE-N/N and sACE-C/C) are not equivalent. The sACE-N/N interface is larger and stronger, burying a surface area of approximately 1,150 Å², compared to a buried surface area of approximately 750 Å² for the sACE-C/C interface.

The interface between sACE-N is formed by a hydrophobic core, anchored by a Y494 π -stacking interaction, and flanked by salt bridges formed by R482 and R487 as well as E241 and E248 along with a collection of polar and van der Waal contacts (**Figure 3B** [↗](#)). The dimer interface between

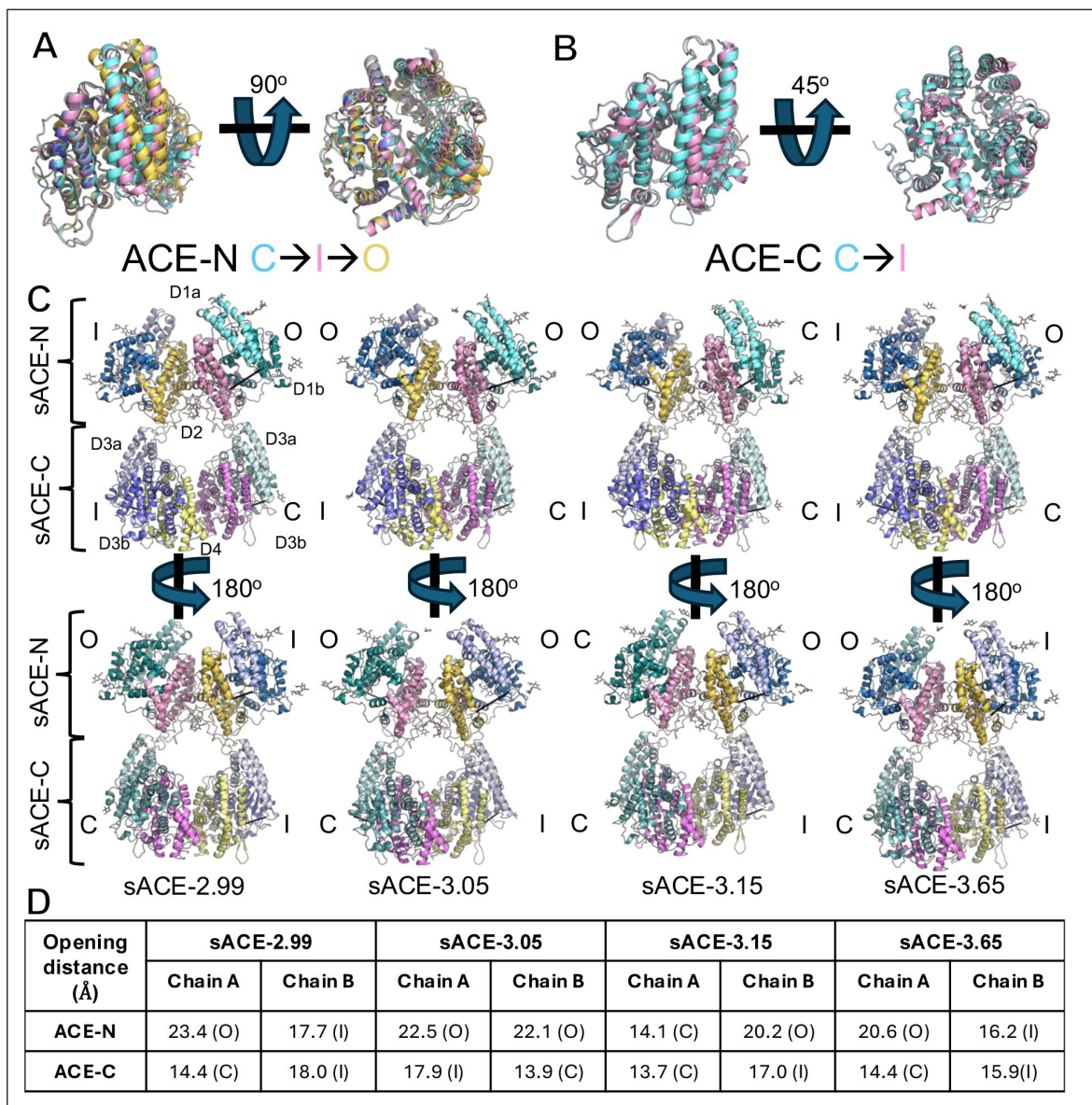


Figure 2

Overall structure of human ACE.

(A) Overlay comparing sACE-N states highlighting the structure differences between the closed (C), intermediate (I), and open (O) states. (B) Overlay comparing sACE-C states highlighting the structure differences between the closed (C), and intermediate (I) states. We define the state based on the distance between the edge of the D2/4 domain bordering the catalytic cleft (residues 150-155 or 750-755) and the tip of the D1/3a region (residues 70-80 or 676-686): closed <15 Å, intermediate >15 Å and <19 Å, open >19 Å. (C) Overall dimer comparisons. (D) Table of openness measurements for each domain per structure.

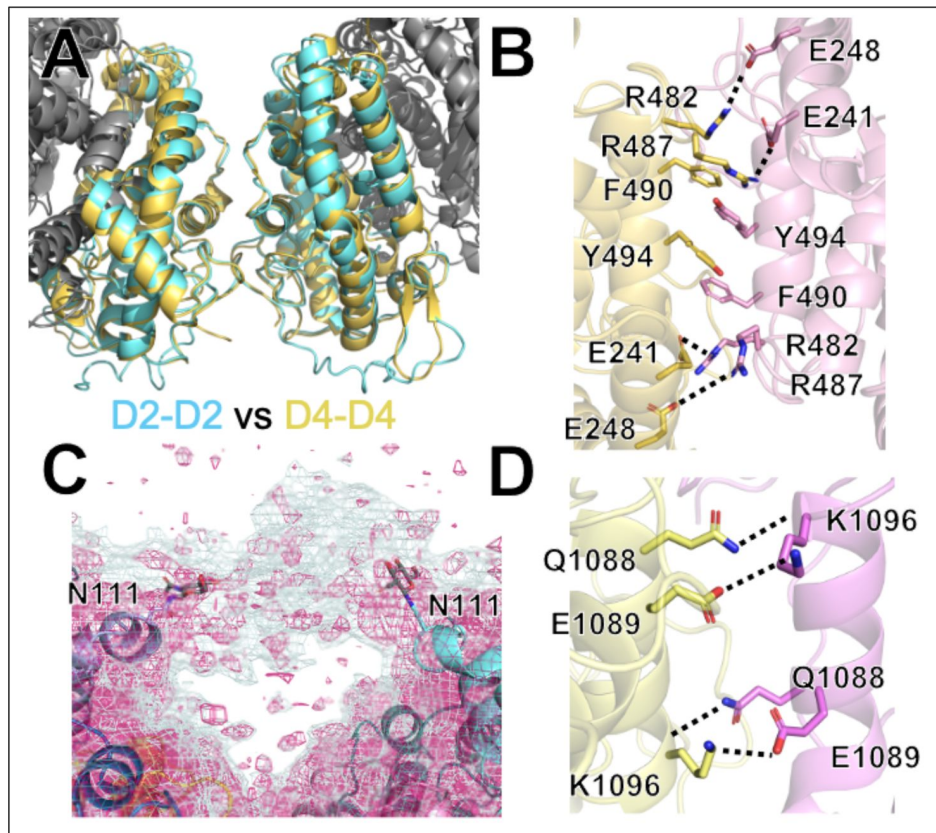


Figure 3

sACE dimerization interfaces.

(A) Overlay comparing sACE-N/N interface (blue) and sACE-C/C (yellow) interfaces. Interfaces adopt the same secondary structure but interacting residues vary between them. (B) Residue-specific interactions at the sACE-N/N interface, see text for details. (C) Unsharpened Coulomb potential density map (cyan) showing density corresponding to glycan-glycan interaction from N111 as part of sACE-N/N interface. Sharpened map is shown in magenta for reference. (D) Residue-specific interactions in the sACE-C/C interface, see text for details.

two sACE-N domains is completed by glycan-glycan interactions between the glycans attached to N111, which is consistent with the role of glycosylation in sACE dimerization (**Figure 3C** [↗](#)) ([41](#) [↗](#)). However, the central residues that form the hydrophobic core of the interface between sACE-N are not present in the interface between sACE-C. Instead, the sACE-C interface is formed by a symmetric salt bridge between K1096 and E1089, side chain-backbone interactions, and van der Waals contacts (**Figure 3D** [↗](#)).

Conformational states of sACE catalytic domains

The comparison of our dimeric sACE cryoEM structures reveals the conformational dynamics of sACE catalytic domains. The four sACE dimer structures represent different combinations of open, intermediate, and closed states of ACE catalytic domains (**Figure 2** [↗](#)). By aligning each sACE-N and sACE-C region in our structures to the D2/4 domain, it becomes clear that our structures represent a continuous gradient of structural heterogeneity, ranging from the most closed state to the most open state (Figure S6).

The comparison of sACE-N in its open and closed states reveals the structural basis for the open-to-closed transition of this domain. Our structures show that when the tip connecting the two long helices of the D1a subdomain moves away from the D2 subdomain, resulting in the open state, the opposite end of the helix moves closer to the D2 subdomain, as if the helix were balanced upon a fulcrum (**Figure 4A, B** [↗](#)). Detailed structural analysis indicates that the fulcrum is primarily formed by the salt bridge between two highly conserved residues: E95 in the D1a subdomain and R137 in the D2 subdomain. To stabilize the open state, additional salt bridges are formed, with D218 in the D2 subdomain pairing with K102 in all cases and occasionally with R125 in the D1a subdomain (**Figure 4A** [↗](#)). The loss of this salt bridge in the closed state is compensated for by the formation of a network of van der Waals interactions between the D1a and D2 subdomains that stabilize the closed state. The transition between open and closed states is likely influenced by the binding of substrates and reaction products during the ACE catalytic cycle. Although a similar fulcrum mechanism is not evident in the sACE-C domain, the connection between the D3a and D2 subdomains would act as a lever to pivot sACE-C from the closed to the open state. The closed state of sACE-C is also stabilized by a network of van der Waals contacts between the D3a and D4 subdomains.

On average, sACE-N is more open than sACE-C, likely due to multiple factors (**Figure 2D** [↗](#)). In sACE-N, the D1a subdomain is at the N-terminal end of sACE, allowing for a greater range of motion (**Figure 4B** [↗](#)), while the fulcrum and stabilization mechanisms can maintain sACE-N in a more widely open state (**Figure 4A** [↗](#)). Conversely, in sACE-C, the motion of the D3a subdomain is constrained by its connection to the D2 subdomain (**Figure 4C** [↗](#)). Additionally, there are sequence differences at the interface between the three subdomains of sACE-N and sACE-C. In the most closed state—the B chain sACE-C domain of the sACE-3.05 structure—residues from all three subdomains, including I678, V753, V955, and V956, form a hydrophobic latch that stabilizes the closed state (**Figure 4D** [↗](#)). The hydrophobic latch in sACE-N is weaker because the residue corresponding to V753 has been replaced with a threonine (**Figure 4D** [↗](#)).

We performed a series of all-atom molecular dynamics (MD) simulations to explore the conformational dynamics of sACE. A total of 8 apo-sACE simulations were conducted, comprising four simulations with non-glycosylated sACE (>1.8 μ s each) and four with sACE where all 12 N-linked glycosylation sites have complex glycans (>2 μ s each) (Movie S5, S6). Glycan orientations were randomized prior to each simulation to avoid bias. We did not observe a substantial difference in the global conformational dynamics between our sets of simulations with and without glycans. The domain motions of these simulations are summarized in Table S2. Below, we focus on the general features from these simulations.

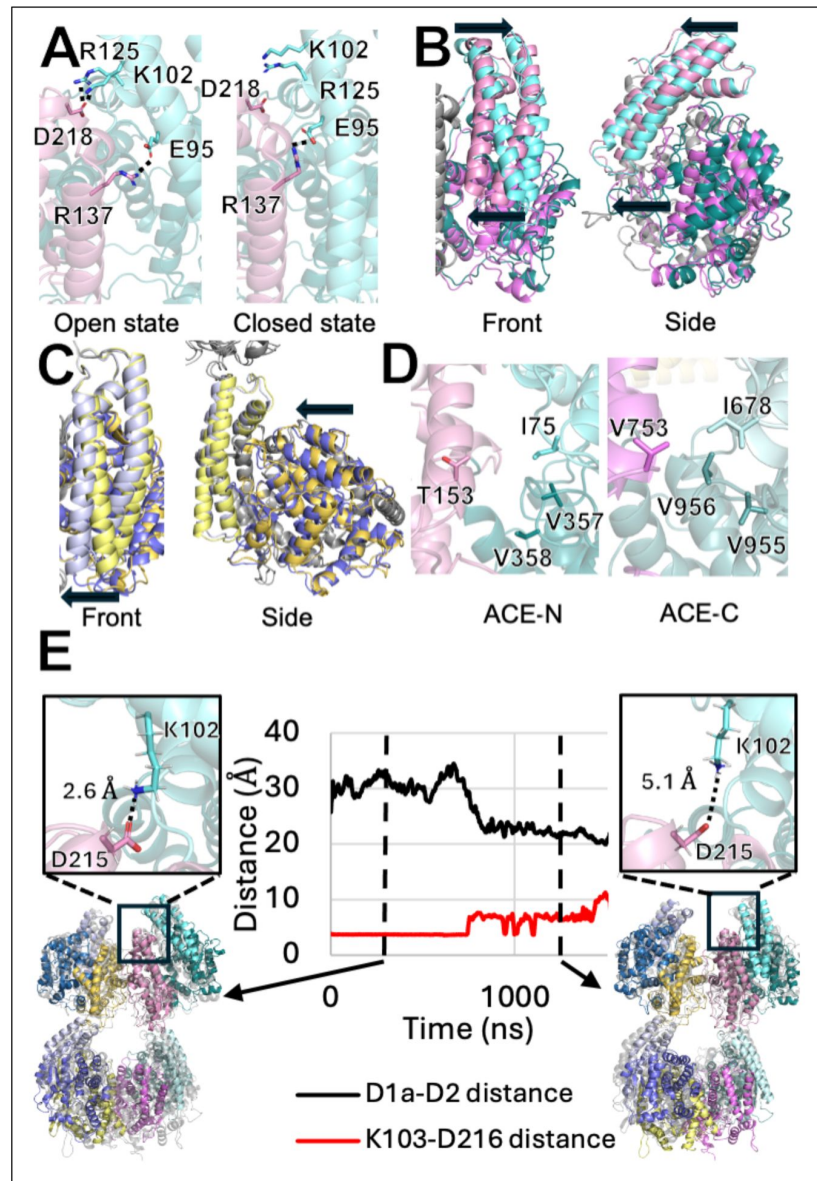


Figure 4

Structural mechanism of the sACE open/close transition.

(A) sACE-N overlay comparing the open (colored in left panel, transparent gray in right panel) and closed (transparent gray in left panel, colored in right panel) states in detail. The open state is stabilized by interaction between residues in the D1a (cyan) and D2 (pink) regions that, notably K102-D218. In the closed state, the K102-D218 interaction is broken. (B) Overlay of the sACE-N open (cyan) and closed (pink) states showing the range of motion. The D1a region rotates about a fulcrum region described in (A), while the D1b region moves as a rigid body. (C) Overlay of sACE-C closed (light blue) and intermediate (yellow) states. Unlike sACE-N, the “top” of the D3a region is constrained by its connection to sACE-N and largely immobile. The primary source of opening is only the motion of the D3a tip. We did not observe any open state structures of sACE-C, suggesting a smaller range of motion relative to sACE-N. (D) Comparison of the hydrophobic “latch” region formed in the closed state between residues of the D1/3a, D1/3b, and D2/D4 domains. V753 in sACE-C has been replaced by T153 in sACE-N, suggesting that the closed state in sACE-N may be less stabilized than the sACE-C closed state. (E) Example all-atom MD simulation tracking the openness of one sACE-N region (black line) and this distance between K102 and D218 (red line). These residues form a salt bridge early in the simulation when sACE-N is open (left inset) but the interaction breaks as sACE-N transitions to the closed state (right inset). Distance measurements for MD simulations were consistently greater than distance values in our static structures and cannot be directly compared to [Figure 2](#).

RMSD analysis reveals that the dominant source of conformational change in our simulations is the open-close transition between the D1/3 and D2/4 domains (Figure S7). The D2/D2 and D4/D4 regions are the most stable, with the bulk of conformational changes localized to the D1/D3 domain. Within the D1/3 domain, we observed that the D1/3a motion was not always correlated with D1/3b or D2/D4, meaning that at a given time, D1/3a could move along with D1/3b, or with D2/D4, or independently. Consistent with our structural analysis, our MD simulations also indicate that sACE-N displays a much wider range of opening geometries than sACE-C. Most of our MD simulations rapidly closed, preventing an effective comparison of the open vs closed state dynamics. However, the slowest closing simulation did reveal a correlation with our previously described “fulcrum” mechanism of opening for sACE-N (Figure 4A). Our simulation revealed that the K102-D215 interaction was maintained for the entire time that sACE-N adopted an open conformation, yet this interaction was lost when sACE-N transitioned to the closed state (Figure 4E).

The MD simulation of glycosylated ACE provides additional structural insights into the structure and functions of N-linked glycosylation of ACE. Glycosylation is shown to affect the folding, secretion, and stability of ACE (42, 43). Our MD simulation reveals that the dynamics of 12 N-linked glycans cover a large surface area of ACE, shielding it from environmental insults (Figure S8). Consistent with observations from our Coulomb potential density maps and previously published data, we observed a substantial and persistent inter-subunit interaction between glycans from N111 (Figure 3C). From room mean square fluctuation (RMSF) analysis, glycosylation was found to increase the overall dynamicism of sACE (Figure S9). Furthermore, while there are N-linked glycans near the catalytic chamber, particularly those at the tip of D1a and D3a, they were highly dynamic and did not occlude access to the catalytic chamber. This is consistent with the observation that deglycosylation only modestly increases the enzymatic activity of ACE (44–46).

We also performed three simulations on the non-glycosylated sACE monomer (>1 microsecond each) (Table S2). We observed the expected intra-subdomain open-closed motions in sACE-N and sACE-C that were also observed in the simulations of ACE dimer. We also observed that the motion of sACE-C relative to sACE-N was significantly greater in our monomer simulations compared to our dimer simulations. This suggests that dimerization places significant constraints upon the conformational dynamics of sACE. Previous work has defined the conformational dynamics of sACE from cryoEM data where both sACE-C domains were not resolved. Given our observations from the monomeric and dimeric MD simulations, we endeavored to understand the conformational heterogeneity present within our cryoEM data to better understand the impact of dimerization on sACE dynamics.

CryoEM heterogeneity analysis of sACE

CryoEM structures are ensemble averages of many aligned particle images. Recently, there have been many approaches developed to understand the particle heterogeneity present within a cryoEM structure as a proxy for molecular motion (47). We employed three methods of heterogeneity analysis: cryoDRGN, RECOVAR, and 3D Variability Analysis (3DVA) (48–50). CryoDRGN employs neural networks to resolve structural heterogeneity in a nonlinear, multidimensional latent space, while RECOVAR and 3DVA both employ linear subspace methods. 3DVA finds the linear subspace by an alternating optimization scheme to capture image variations while RECOVAR utilizes a regularized covariance estimator to identify subspaces that best capture the distribution of states.

We conducted this analysis on both of our datasets and found the results to agree, therefore, we focus on the larger, and higher-resolution dataset collected at NCCAT below. For analysis, the entire particle set, prior to 3D classification and consisting of ~170k particles, was utilized. The particle alignments from the same C1 consensus refinement were used as input for each of the three methods as current methods to investigate particle heterogeneity are highly dependent upon

initial pose assignment of individual particles. Globally, all three methods agreed with our ensemble structures, in that the dominant source of structural variability within our particle population could be described by the open-closed transition of individual domains, and that sACE-N displayed more variability than sACE-C.

Initially, we performed 3D variability analysis (3DVA) in cryoSPARC to provide a direct comparison to the previously published analysis of the partial ACE cryoEM structure (51 [DOI](#)) (**Figure 5A** [DOI](#)). Trajectories were reconstructed along the top 5 principal components of variance (Movie S7). Surprisingly, the trajectory describing the variance along the first principal component vector suggests a dominant inter-domain motion, where the sACE-N/N region bends towards the sACE-C/C region. The trajectories along the remaining principal component vectors comprise varying combinations of individual domains undergoing the open-close transition. Consistent with our previous observations, the range of presumptive motion in these trajectories is greater in sACE-N than sACE-C. Previous 3DVA performed on the ACE monomer structure revealed 4 types of potential motion, termed bending, pivoting, breathing, and jumping (28 [DOI](#)). The breathing motion matches the open-close transition that dominates most of our principal component vectors, while the bending motion appears to correlate with the variance described by our top principal component vector. We did not observe any potential motions reminiscent of the jumping or pivoting motions in our 3DVA. Furthermore, the magnitude of the bending motion in our 3DVA is substantially less than previously reported for the partial dimeric sACE structure, which likely suffered from substantial denaturation (28 [DOI](#)). Together, this highlights a significant difference in the conformational dynamics between sACE dimer and previously reported cryoEM structures. This type of analysis is sensitive to artifacts introduced by particle damage and denaturation, underscoring the importance of preserving protein integrity via sample preparation and the use of Chameleon to reduce the exposure to air-water interface. This also suggests dimerization functions as a key constraint to the conformational dynamics of sACE.

We then moved on to analyze our data using cryoDRGN, which revealed that the data formed a single cluster across multiple dimensions (**Figure 5B** [DOI](#), S10). Trajectories generated along five principal component vectors of structural variance were dominated by the open-close transition, primarily in sACE-N (Movie S8). We observed little variance in sACE-C and, unlike 3DVA, cryoDRGN did not reveal any substantial inter-domain motions. Faced with this discrepancy between the 3DVA and cryoDRGN results, we employed a third method, RECOVAR, to further analyze our data. Similar to cryoDRGN, RECOVAR revealed that our data formed a singular cluster across multiple dimensions of analysis. Volumes representative of kmeans cluster reconstructions were similarly consistent with our cryoDRGN results. RECOVAR indicated that the primary source of structural variance could be explained by the open-close transition, primarily of the sACE-N domains. With this in mind, we repeated the RECOVAR analysis with a focus mask around the sACE-N/N region. We then estimated the conformational density by deconvoluting the multidimensional latent space with a two-dimensional principal component analysis. The resulting density plot revealed four clusters which corresponded to different conformations of the sACE-N/N region (**Figure 5C** [DOI](#)). The most populous cluster corresponded to a state where both sACE-N subunits adopted an open conformation. The second largest cluster, and the smallest cluster, corresponded to a state where one sACE-N was open and the other was closed. The discrepancy in population size between these clusters is likely due to bias in the initial particle poses, rather than a subunit-specific preference for the open state. The final cluster corresponded to a state where both sACE-N subunits adopted the closed conformation. We then used RECOVAR to generate a trajectory through the latent space to predict a potential transition pathway from the closed-closed state to the open-open state (Movie S9). This trajectory revealed that one subunit would open, followed by the second, and raised the question of potential allostery or coordinated motion between the adjacent sACE-N domains. 3DVA, cryoDRGN, and RECOVAR all appeared to display varying degrees of coordinated motion between sACE domains, yet this type of analysis lacks an explicit time component, so it is not possible to draw concrete conclusions regarding coordinated motion. Likewise, while MD simulations contain an explicit time factor, most of our MD simulations closed quickly, and did not

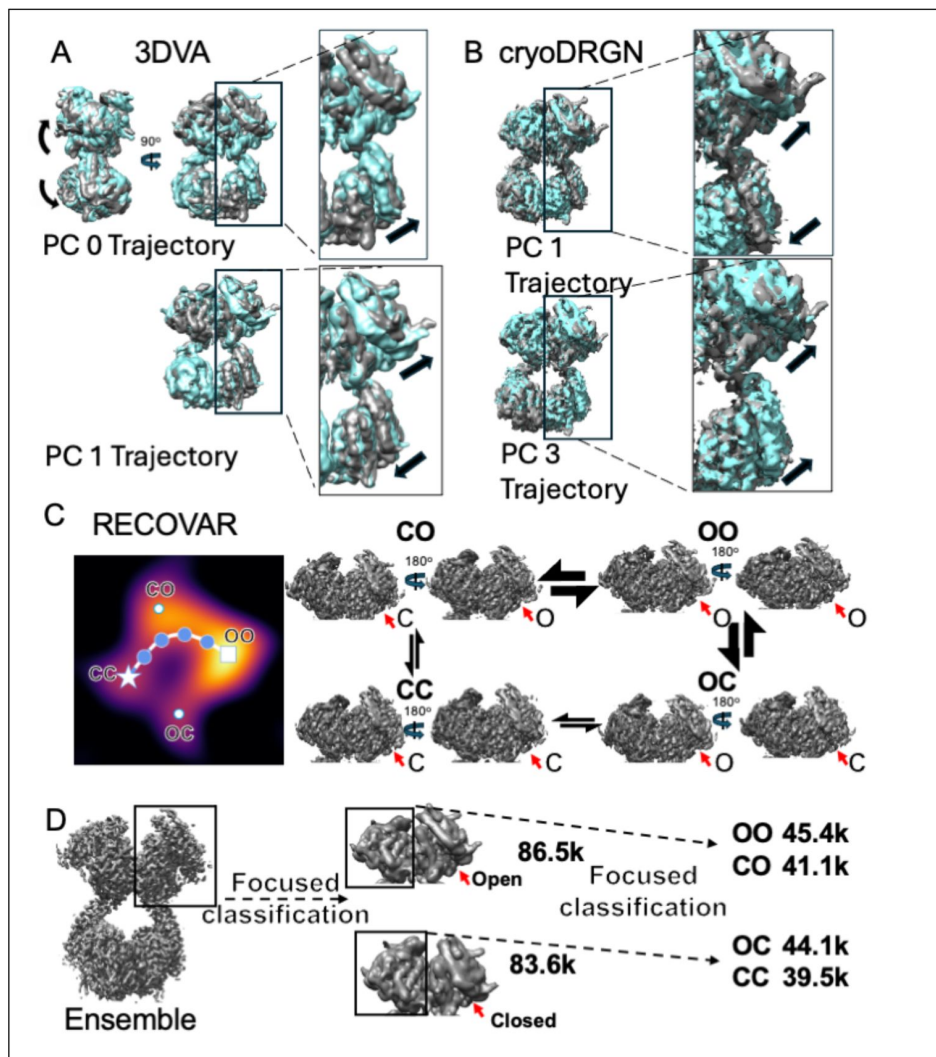


Figure 5

CryoEM heterogeneity analysis.

(A) Visualization of the structural changes revealed by cryoSPARC 3DVA trajectories calculated along two principal components (PCs) of structural variance. Starting states are shown in cyan, ending states in gray. PC 0 reveals a large, inter-domain bending motion accompanied by the open/close transition in sACE-C. PC 1 and the remaining PCs are dominated by the open/close transition of individual regions. See Table S3, and Movie S7 for additional details. (B) Visualization of the structural changes revealed by the cryoDRGN trajectories calculated along two PCs of structural variance. Starting states are shown in cyan, ending states in gray. PCs are dominated by the open/close transition of individual regions. See Table S3, and Movie S8 for additional details. (C) Analysis in RECOVAR with a focus mask on sACE-N/N reveals that particles adopt roughly four clusters within the latent space (heat map of particle density) corresponding to the open-open (OO, white square), open-closed (OC, white dot), closed-open (CO, white dot) and closed-closed (CC, white star) states of sACE-N/N. A trajectory estimating the path through latent space corresponding to the structural transition from sACE-N/N CC state to the OO state (blue points) suggests that individual sACE regions transition at different rates. See Movie S9 for trajectory. (D) Focused 3D classification was performed in cryoSPARC to explore evidence of coordinated motion between sACE-N regions. 3D classification focusing on one sACE-N region revealed 2 roughly equal classes of particles: open and closed. Subsequent 3D classification focused on the other sACE-N region again revealed 2 roughly equal classes, suggesting the lack of coordinated motion between sACE-N regions in the absence of substrate.

reopen. Thus, while we did not observe coordinated motion from our simulations, no conclusion can be reached because of limited intradomain motions. To investigate coordination between the sACE-N domains, we performed 3D classification in cryoSPARC with a focus mask on a single sACE-N domain and requested 2 classes (**Figure 5D** [↗](#)). The particles were split roughly evenly between the two classes, one resulting in an open state and the other being closed. We then used these classes for another round of 3D classification with a focus mask on the opposite sACE-N domain and again asked for 2 classes. Again, these particles were split roughly evenly among the classes, leaving us with four similarly sized classes adopting the same conformations as the clusters revealed by RECOVER. As a result, we are unable to find any evidence of coordinated motion between sACE-N domains in the absence of substrate.

Discussion

Our cryoEM and MD simulations analysis of the extracellular regions of dimeric ACE offers insights into the stability of ACE catalytic domains. We show that ACE dimer interface consists of three major components, protein-protein interfaces between ACE-N, glycan-glycan interaction between ACE-N, and protein-protein interface between ACE-C. The interface between ACE-N, in general, is 50% larger and has higher numbers of hydrogen bonds and salt bridges than that between ACE-C. The necessity for the stronger interaction between ACE-N is likely due to that ACE-N is distal to transmembrane helix so that the additional reinforcement is required. The weaker interaction and lack in the contribution from glycan in the interface between ACE-C could also explain the observed preferential denaturation of ACE-C despite the high structural similarity between ACE-N and ACE-C. The denaturation is likely due to the exposure to air-water interface during vitrification, a common issue in single particle cryoEM analysis ([35](#) [↗](#), [36](#) [↗](#)). We have partially overcome the denaturation issue of ACE-C by optimizing the buffer pH to increase the melting temperature and using the faster plunging time during vitrification afforded by Chameleon to reduce the exposure to air-water interface (**Figure 1** [↗](#)). However, we did not completely prevent the preferential denature of ACE-C during vitrification.

ACE is part of a family of proteases capable of effectively degrading A β . This family includes M16 metalloproteases like IDE and PreP, and M13 metalloproteases such as neprilysin and endothelin converting enzyme. Integrative structural analysis has demonstrated how IDE employs conformational dynamics, charge, and secondary structure complementarity to selectively capture peptide substrates into its sizable catalytic chamber, subsequently unfolding and degrading amyloidogenic peptides such as insulin and A β , without degrading structurally similar yet non-amyloidogenic peptides ([25](#) [↗](#), [26](#) [↗](#), [52](#) [↗](#)). The other mentioned enzymes also belong to this group of chamber-containing proteases, known as cryptidases ([24](#) [↗](#), [25](#) [↗](#), [53](#) [↗](#)). Similar to cryptidases, structural analysis reveals that ACE must undergo an open-closed transition to capture, engulf, and degrade its oligopeptide substrates, such as angiotensin 1 and bradykinin.

Our apo-ACE structures and MD simulations identify a fulcrum mechanism that governs the open-closed transition of ACE-N, while the interplay between ACE-N and the D3a domain likely plays a crucial role in this transition. Analogous to the binding of MB60, a PreP inhibitor, to the hydrophobic pocket to trigger the open to closed transition, factors such as binding of substrates and ACE inhibitors may alter the balance, leading to the regulation of ACE activity ([24](#) [↗](#)). We also observe that the catalytic pocket of the ACE catalytic domain is not large enough to engulf the entire A β . Additionally, the distances between the domains (ACE-N to ACE-C, ACE-N to ACE-N, or ACE-C to ACE-C) are too large for these domains to bind A β simultaneously ([54](#) [↗](#)). Therefore, ACE likely employs a different mechanism from IDE and PreP to selectively bind and degrade A β . Future integrative structural studies await to reveal how ACE selectively recognizes and degrades a diverse array of substrates.

Single particle cryoEM holds significant promise for revealing the conformational dynamics of biological macromolecules from embedded heterogeneity. Consequently, cryoEM heterogeneity analysis is an active and rapidly evolving field providing insights into the conformational dynamics and allostery of macromolecules (47 [↗](#)). However, many challenges exist in using cryoEM heterogeneity analysis due, in part, to the low signal-to-noise ratio in observed micrographs. Often, these analyses address questions difficult or impossible to tackle experimentally, leaving no ground truth for validation. We encounter such challenges in our heterogeneity analysis as well. Consistent with our single particle cryoEM structural analysis, we observed the expected intradomain open-closed transition using cryoDRGN, cryoSPARC 3DVA, RECOVER, and all-atom MD simulation. Surprisingly, we did not observe such motion using cryoSPARC 3DFlex, a neural network-based method analyzing our cryoEM data of sACE (48 [↗](#)). Central to the working of cryoSPARC 3DFlex is the generation of a tetrahedral mesh used to calculate deformations within the particle population. Proper generation of the mesh is critical for obtaining useful results and must often be determined empirically. Despite several attempts, we were unable to obtain mesh conditions suitable to observe any open-close transition, illustrating a limitation of cryoSPARC 3DFlex.

The interplay between ACE-N and ACE-C within the ACE dimer has complicated consequences for ACE functions, showing negative cooperativity for enzymatic activities while exhibiting positive synergy in A β cleavage by ACE-N and activity at the bradykinin receptor (11 [↗](#), 31 [↗](#)–34 [↗](#)). Understanding the conformational dynamics in both inter- and intra-domain motions of the extracellular regions of the ACE dimer holds promise for designing ACE modulators to improve and broaden the therapeutic use of ACE modulation. Our cryoEM heterogeneity analysis has revealed coordinated open-closed transitions across four catalytic domains using cryoDRGN and cryoSPARC 3DVA, which could potentially provide the molecular basis for ACE allostery (**Figure 5A,B** [↗](#), Movies S7,8). However, our RECOVER analysis reveals independent motion of two ACE-N domains, not the coordinated motion revealed by cryoDRGN and cryoSPARC 3DVA (**Figure 5C** [↗](#)). Furthermore, our focused classification using cryoSPARC failed to reveal a profound association between specific conformational states of the ACE catalytic domains, an expectation from the trajectories revealed by cryoDRGN and cryoSPARC 3DVA (**Figure 5D** [↗](#)).

The first principal component trajectory in our 3DVA analysis revealed a significant potential interdomain motion between sACE-N/N and sACE-C/C, yet cryoDRGN and RECOVER did not reveal any inferred motions of similar magnitude. Notably, the trajectory revealing this potential interdomain motion in 3DVA was absent any variability in the sACE-N open/close transition. In our study, this conformational heterogeneity appears to stem from subtle yet significant conformational differences between the two chains within an ACE dimer and the inability of cryoSPARC to properly align these conformationally distinct chains (Figure S11). CryoDRGN and RECOVER revealed smaller magnitudes of potential interdomain motions, which were often accompanied by various combinations of individual domains opening and/or closing. This suggests that the apparent discrepancy in the magnitude of inferred interdomain motion could be a result of the different theoretical approaches of dimensionality reduction and trajectory generation employed by the respective methods of analysis and underscores the importance of incorporating multiple methods of analysis in the absence of a singular “gold-standard” of heterogeneity analysis.

Advances in single-particle cryoEM analysis have enabled detailed structural examination of the extracellular domains of the ACE dimer. Beyond its implications for future drug discovery efforts aimed at enhancing ACE-based therapies, ACE can serve as a valuable model of a multi-domain glycoprotein for studying methods to preserve sample integrity, particularly due to the high propensity of ACE-C to denature during vitrification. Furthermore, considering the aforementioned challenges and caveats, we propose that, due to its size, biomedical importance, and the complexity of its motions, ACE also represents an excellent model system for evaluating and improving cryoEM heterogeneity analysis.

Experimental procedures

Expression of human sACE

Human 293T and 293F cells were maintained in DMEM + 10% fetal calf serum (FCS). The lentiviral plasmid for the expression of the extracellular domain of human ACE (sACE, aa 1-1231) that contains a 6-histidine tag at its C-terminus was made by Vectorbuilder and co-transfected with helper plasmids to human 293T cells for lentiviral production. Human 293F cells were infected with the resulting virus and selected for stable sACE expression. 293F that stably expressed sACE grew poorly in Free-style 293 media (ThermoFisher 12338018). Thus, for ACE production, 293F cells that stably expressed sACE were first grown in DMEM+10% FCS at 37°C and gradually shifted into Free-style 293 media by reducing FCS by first in Free-style 293 media + 2% FCS and then to media without FCS. These cells were then shifted to and maintained in 30°C for 3-5 days for ACE production to reduce proteolysis.

Purification of soluble human ACE

Culture supernatant was centrifuged at 3,000xg for 5 min, diluted 10-fold with 25 mM Tris-HCl pH 8, filtered, and run over a Source Q column pre-equilibrated with 25 mM Tris-HCl pH 8. Column was washed to baseline with Tris-HCl pH 8 and bound protein was eluted with a linear gradient of NaCl to 1 M. sACE containing fractions were pooled and EDTA was added to 10 mM to strip the catalytic zinc ion. Sample was concentrated and run over a Superdex S200 SEC column pre-equilibrated with the desired cryoEM buffer.

Differential scanning fluorometry (DSF)

To optimize the grid making conditions, differential scanning fluorimetry was applied to screen 96 buffers conditions. DSF was performed with 3 mg/ml sACE and 10X Sypro Orange in 20 µl buffers in a Thermo Fisher Step ONE RT-PCR thermocycler. Using the melting temperature and slope as selective criteria, 25 mM citrate pH 5.5, 25 mM sodium phosphate pH 5.5, 150 mM NaCl, 10 mM EDTA was identified as the ideal buffer condition and used for all subsequent cryoEM experiments.

CryoEM data collection and processing

Peak fractions were collected off the SEC column and used to immediately prepare grids, resulting in a protein concentration of ~4.5 mg/ml. Quantifoil R1.2/1.3 Cu200 grids were glow-discharged in air and grids were prepared by Vitrobot mark 3. 3.5 µl of sample was applied to the grid in the chamber with 100% humidity, 25°C. Excess sample was blotted for 1 sec, blot force 2, and grids were plunge frozen in liquid ethane. All images were acquired using a Titan Krios microscope (FEI) operated at 300KeV with a Gatan K3 direct electron detector. A dataset of ~3,600 micrographs was collected at The University of Chicago Advanced Electron Microscopy facility. The dataset was processed using cryoSPARC (Figure S1) (55 [DOI](#)). Briefly, a four-domain sACE dimer model was generated using sACE-N dimer and sACE monomer and adjusting the distance between ACE-C domains to avoid steric clashes as a template for particle picking (28 [DOI](#)). After template picking, 1.1 million particles were extracted with a box size of 360 pixels. Several rounds of 2D classification were used to remove junk particles. ~577K particles were processed in successive rounds of *ab initio* reconstruction and heterogeneous refinement. sACE grids were also prepared using Chameleon and collected a dataset of ~19,000 micrographs at the National Center for CryoEM Access and Training. For those, Quantifoil Active R 1.6/0.9 Cu/Rh 300 were coated with gold using a Safematic CCU-010 Compact Coating Unit. Briefly, grids were placed nanowire side down on a glass slide, gold disks were sputtered with a 35 nm target thickness at 1.5×10^{-2} Torr, and a process current of 30 mA. Grids were glow discharged with air at 12mA for 80 secs using the

internal Chameleon glow discharger. The grids were plunged at 160 to 180 milliseconds. All images were acquired using a Titan Krios microscope (FEI) operated at 300KeV with a Gatan K3 direct electron detector (Gatan). Images were automatically acquired using Leginon ([56](#)) using collection parameters as shown in Table S1. This dataset was processed in cryoSPARC (Figure S2). Briefly, templates were generated from our full-length ACE structure described above using vitrobot-prepared grids. Micrographs with a max CTF value worse than 4Å were removed from the dataset, leaving ~12,000 micrographs for template picking, which yielded 10.7 million particles. The particle set was trimmed to ~2.8 million based on quality metrics and run through 2 rounds of 2D classification, reducing the particle number to ~660,000. 2D classes were dominated by 4 domain views, and yielded a much greater diversity of views, indicating that Chameleon preparation produced a dramatic improvement in particle quality. Initial 3D homogeneous refinement reached 3.6 Å, matching the previous final structure. Four maps from 3D classes that contains all four catalytic domains of ACE, one from particles derived from grids prepared by using vitrobot and three from those prepared by using Chameleon, were used for structure building using Chimera and Coot and the structure was refined using Phenix real space refinement. Structural analysis was performed using PYMOL and Chimera.

Heterogeneity analysis

All heterogeneity analysis was performed on the full final particle set, encompassing all of the NCCAT structures. Initially, cryoSPARC 3DVA was performed on both the NCCAT- and UChicago-collected datasets and results were found to be consistent, so the NCCAT-collected dataset was used for further analysis, given its higher particle number and quality. Input particle poses for each method of heterogeneity analysis were derived from the same C1 refinement job in cryoSPARC, which reached 2.88 Å. 3DVA was used to solve the top 5 modes of variance with a filter resolution of 6 Å. Particles were downsampled ~3x to a box size of 128 pixels in cryoDRGN to speed up processing. The same downsampled particle set was used for both cryoDRGN and RECOVER. CryoDRGN was run with default parameters and the analysis was performed on the top 5 principal components for direct comparison to cryoSPARC 3DVA. RECOVER pipeline.py was initially run on the full sACE structure. Reconstructions representative of kmeans cluster centers revealed the dominant source of structural variability to be the open-close transition in sACE-N. As a result, RECOVER was run again with a focus mask on sACE-N/N and a zdim=4. Conformational density was estimated with a 2-dimensional PCA, and an index value of 3 was chosen for further analysis. The points in latent space representing the top 12 stable conformational states were estimated from the top 20% of particles and representative volumes were generated. The generated states populated four clusters and the 12 states were simplified to 4 distinct conformations where sACE-N/N adopted the open-open (OO), open-closed (OC), closed-open (CO), and closed-closed (CC) states (Figure 5C). We then generated a trajectory through the latent space using the OO and CC representative volumes as endpoints (Movie S7).

All-atom MD simulations and analysis

The sACE-3.65 structure was used to initiate all MD simulations. For the glycosylated sACE simulations, N-linked glycans were built onto all 12 known N-linked glycosylation sites (residues 38, 54, 74, 111, 146, 318, 445, 509, 677, 695, 714, 760) with the complex glycan BGLC-BGLC-BMAN-AMAN-BGLC-ANE5-AMAN-BGLC-BGAL-ANE5-AFUC using CHARMM-GUI ([57](#)). Simulations were prepared in QwikMD or CHARMM-GUI and run at pH 7, 150 mM NaCl, 310 K, 1 atm with periodic boundary conditions in NAMD3.0 using the University of Chicago Beagle3 partition ([58](#)). Explicit solvent was described with the TIP3P model ([59](#)). Data was analyzed in VMD 1.9.4 ([60](#)).

Data availability

Unprocessed micrographs, particle stacks, and associated pose/CTF information have been deposited to EMPIAR under the accession numbers: EMPIAR-12181 (Vitrobot-prepared grids) and EMPIAR-12484 (Chameleon-prepared grids)

EM structures and maps have been deposited to the EMDB and PDB, respectively, under the following accession numbers:

- sACE-2.99: 9D5S and EMD-46581
- sACE-3.05: 9D5M and EMD-46579
- sACE-3.15: 9D55 and EMD-46574
- sACE-3.65: 9CLX and EMD-45733

Acknowledgements

We are grateful to Mahira Aragon for grid preparation using Chameleon, Kasahun Neselu for data collection, and Marc Auréle Gilles for assistance with RECOVAR analysis. This work was supported by the NIH grant GM121964 to W.-J. Tang. Some of this work was performed at the National Center for CryoEM Access and Training (NCCAT) and the Simons Electron Microscopy Center located at the New York Structural Biology Center, supported by the NIH Common Fund Transformative High Resolution Cryo-Electron Microscopy program (U24 GM129539,) and by grants from the Simons Foundation (SF349247) and NY State Assembly.

Additional information

Author contributions

J.M.M. and W.-J.T., designed the project. X. W. transduced and selected ACE-expression HEK293F cells. J.M.M. and W.-J. T. performed the expression and purification of ACE. J.M.M. performed cryoEM grid preparation, data acquisition and processing assisted by M.Z. and overseen by W.-J. T., and M. Z. J.M.M. built and refined cryoEM structural models and performed heterogeneity analysis. J.M.M. and W.-J. T. performed all-atom MD simulations and analysis. J.M.M. and W.-J.T. wrote the manuscript and all authors contributed to manuscript finalization.

Additional files

Supplemental figures+tables+movie legends [↗](#)

Movie S1 [↗](#)

Movie S2 [↗](#)

Movie S3 [↗](#)

Movie S4 [↗](#)

Movie S5 [🔗](#)

Movie S6 [🔗](#)

Movie S7 [🔗](#)

Movie S8 [🔗](#)

Movie S9 [🔗](#)

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Reviewer #1 (Public review):

Summary:

The authors report four cryoEM structures (2.99 to 3.65 Å resolution) of the 180 kDa, full-length, glycosylated, soluble Angiotensin-I converting enzyme (sACE) dimer, with two homologous catalytic domains at the N- and C-terminal ends (ACE-N and ACE-C). ACE is a protease capable of effectively degrading A β . The four structures are C2 pseudo-symmetric homodimers and provide insight into sACE dimerization. These structures were obtained using discrete classification in cryoSPARC and show different combinations of open, intermediate, and closed states of the catalytic domains, resulting in varying degrees of solvent accessibility to the active sites.

To deepen the understanding of the gradient of heterogeneity (from closed to open states) observed with discrete classification, the authors performed all-atom MD simulations and continuous conformational analysis of cryo-EM data using cryoSPARC 3DVA, cryoDRGN, and RECOVER. cryoDRGN and cryoSPARC 3DVA revealed coordinated open-closed transitions across four catalytic domains, whereas RECOVER revealed independent motion of two ACE-N domains, also observed with cryoSPARC-focused classification. The authors suggest that the discrepancy in the results of the different methods for continuous conformational analysis in cryo-EM could result from different approaches used for dimensionality reduction and trajectory generation in these methods.

Strengths:

This is an important study that shows, for the first time, the structure and the snapshots of the dynamics of the full-length sACE dimer. Moreover, the study highlights the importance of combining insights from different cryo-EM methods that address questions difficult or impossible to tackle experimentally while lacking ground truth for validation.

Weaknesses:

The open, closed, and intermediate states of ACE-N and ACE-C in the four cryo-EM structures from discrete classification were designated quantitatively (based on measured atomic distances on the models fitted into cryo-EM maps, Figure 2D). Unfortunately, atomic models were not fitted into cryo-EM maps obtained with cryoSPARC 3DVA, cryoDRGN, and RECOVER, and the open/closed states in these cases were designated based on qualitative analysis. As the authors clearly pointed out, there are many other methods for continuous conformational heterogeneity analysis in cryo-EM. Among these methods, some allow analyzing particle images in terms of atomic models, like MDSPACE (Vuilleminot et al., J. Mol. Biol. 2023, 435:167951), which result in one atomic model per particle image and can help in analyzing cooperativity of domain motions through measuring atomic distances or angular differences between different domains (Valimehr et al., Int. J. Mol. Sci. 2024, 25: 3371). This could be discussed in the article.

<https://doi.org/10.7554/eLife.106044.1.sa2>

Reviewer #2 (Public review):

Summary:

The manuscript presents a valuable contribution to the field of ACE structural biology and dynamics by providing the first complete full-length dimeric ACE structure in four distinct states. The study integrates cryo-EM and molecular dynamics simulations to offer important insights into ACE dynamics. The depth of analysis is commendable, and the combination of structural and computational approaches enhances our understanding of the protein's conformational landscape. However, the strength of evidence supporting the conclusions needs refinement, particularly in defining key terms, improving structural validation, and ensuring consistency in data analysis. Addressing these points through major revisions will significantly improve the clarity, rigor, and accessibility of the study to a broader audience, allowing it to make a stronger impact in the field.

Strengths:

The integration of cryo-EM and MD simulations provides valuable insights into ACE dynamics, showcasing the authors' commitment to exploring complex aspects of protein structure and function. This is a commendable effort, and the depth of analysis is appreciated.

Weaknesses:

Several aspects of the manuscript require further refinement to improve clarity and scientific rigor as detailed in my recommendations for the authors.

<https://doi.org/10.7554/eLife.106044.1.sa1>

Reviewer #3 (Public review):

Summary:

Mancl et al. report four Cryo-EM structures of glycosylated and soluble Angiotensin-I converting enzyme (sACE) dimer. This moves forward the structural understanding of ACE, as previous analysis yielded partially denatured or individual ACE domains. By performing a heterogeneity analysis, the authors identify three structural conformations (open, intermediate open, and closed) that define the openness of the catalytic chamber and structural features governing the dimerization interface. They show that the dimer interface

of soluble ACE consists of an N-terminal glycan and protein-protein interaction region, as well as C-terminal protein-protein interactions. Further heterogeneity mining and all-atom molecular dynamic simulations show structural rearrangements that lead to the opening and closing of the catalytic pocket, which could explain how ACE binds its substrate. These studies could contribute to future drug design targeting the active site or dimerization interface of ACE.

Strengths:

The authors make significant efforts to address ACE denaturation on cryo-EM grids, testing various buffers and grid preparation techniques. These strategies successfully reduce denaturation and greatly enhance the quality of the structural analysis. The integration of cryoDRGN, 3DVA, RECOVAR, and all-atom simulations for heterogeneity analysis proves to be a powerful approach, further strengthening the overall experimental methodology.

Weaknesses:

In general, the findings are supported by experimental data, but some experimental details and approaches could be improved. For example, CryoDRGN analysis is limited to the top 5 PCA components for ease of comparison with cryoSPARC 3DVA, but wouldn't an expansion to more components with CryoDRGN potentially identify further conformational states? The authors also say that they performed heterogeneity analysis on both datasets but only show data for one. The results for the first dataset should be shown and can be included in supplementary figures. In addition, the authors mention that they were not successful in performing cryoSPARC 3DFlex analysis, but they do not show their data or describe the conditions they used in the methods section. These data should be added and clearly described in the experimental section.

Some cryo-EM data processing details are missing. Please add local resolution maps, box sizes, and Euler angle distributions and reference the initial PDB model used for model building.

<https://doi.org/10.7554/eLife.106044.1.sa0>